Angle in a Semicircle

Exam code: 0607

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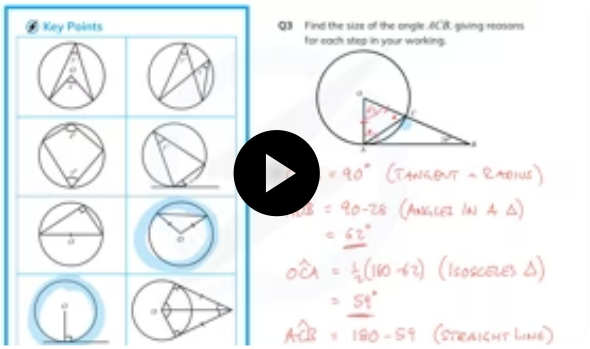

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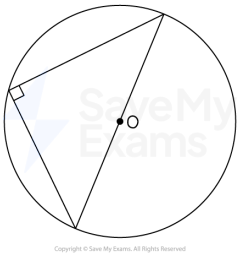
Angle in a semicircle

Circle Theorem: The angle in a semicircle is 90°

Help me understand

- The lines drawn from a **point** on the **circumference** to either end of a **diameter** are **perpendicular**
  - The angle at that point on the circumference is **90°**
  - This circle theorem only uses half of the circle
    - The right-angle is called the angle in a **semicircle**
- This is a **special case** of the circle theorem "the angle at the centre is twice the angle at the circumference"
  - The angle on the diameter is 180°
  - The angle at the circumference is halved, giving 90°



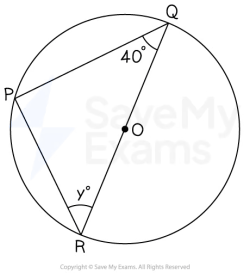
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- To spot this circle theorem on a diagram look for a triangle where

- **all three vertices** are on the **circumference**
- The 90° angle will always be the angle **opposite** the diameter
- When explaining this theorem in an exam you must use the **keywords:**
  - **The angle in a semicircle is 90°**

- Questions that use this theorem may
  - appear in **whole circles** or in **semicircles**
  - require the use of **Pythagoras' Theorem** to find a missing length

Worked Example



P, *Q* and *R* are points on a circle.
RQ is a diameter.

Find the value of y.

Give a reason for your answer.

Answer:

Use the fact that angles in a triangle add up to 180° and the circle theorem

The angle in a semicircle is 90°

Write an equation for y

$$y + 90 + 40 = 180$$

Solve for y

$$y = 180 - 90 - 40$$
$$y = 50$$

$$y = 50$$

The angle in a semicircle is 90°

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